

Sorted Set

Set vs Sorted Set

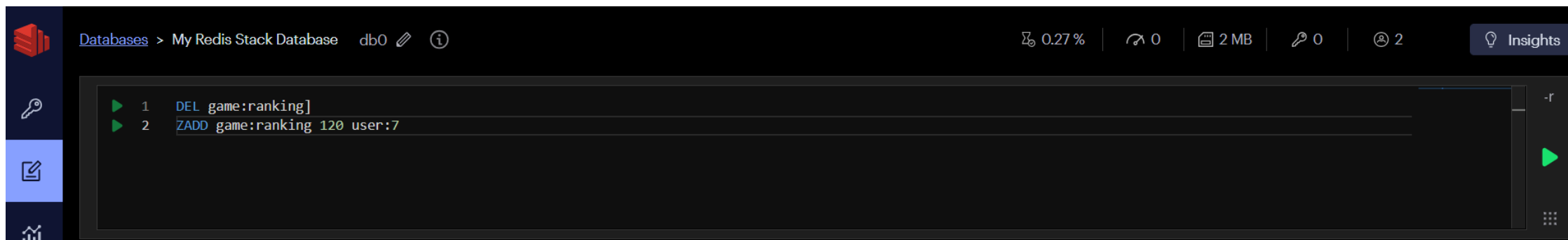
항목	Set (셋)	Sorted Set (정렬된 셋)
한 줄 설명	중복 없는 모음	점수 기준으로 정렬된 모음
중복 값	불허	불허
순서	없음	있음 (score 기준 자동 정렬)
추가 정보	값만 저장	값 + score(점수) 함께 저장
비유	출석부	성적순 출석부
언제 쓰나?	중복 제거가 필요할 때	순위, 랭킹이 필요할 때
대표 명령	SADD SMEMBERS	ZADD ZRANGE ZRANK

1. 점수 저장과 조회

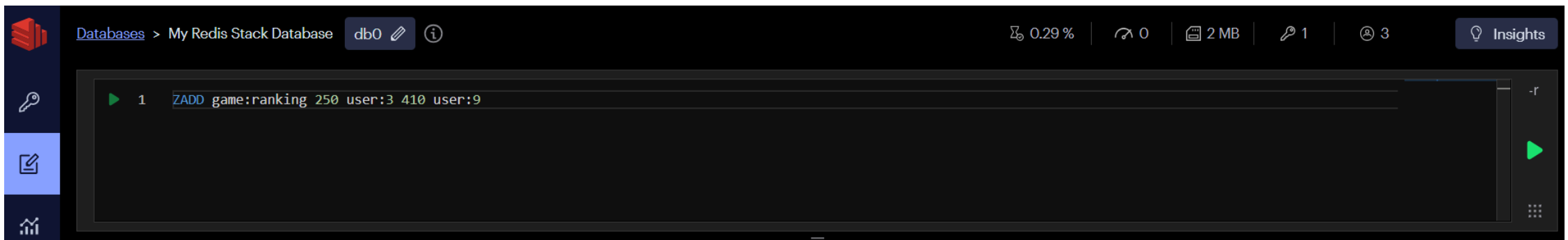
- **ZADD** : 멤버와 점수 추가 또는 변경

```
DEL game:ranking  
ZADD game:ranking 120 user:7
```

```
# game:ranking 키 삭제 (초기화)  
# user7을 점수 120으로 추가
```



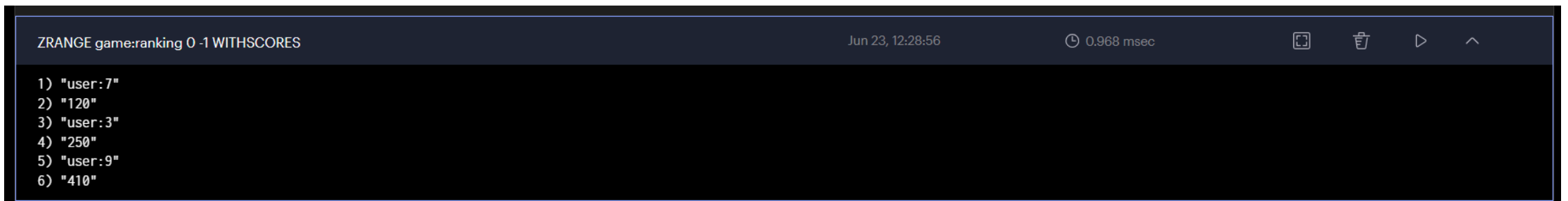
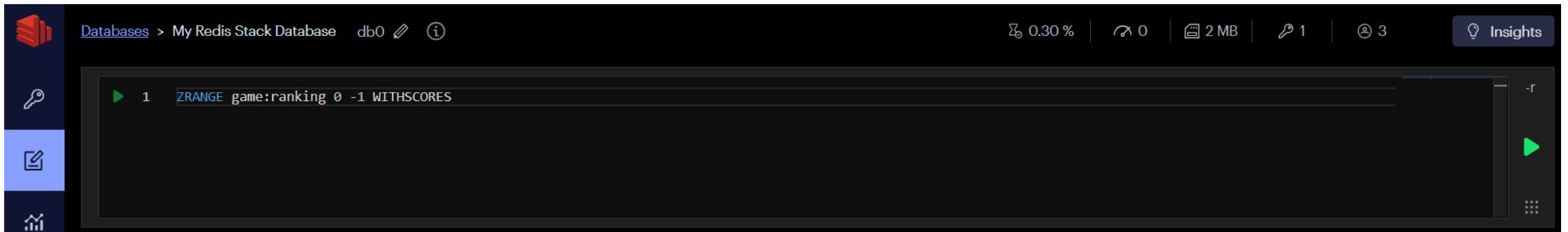
ZADD game:ranking 250 user:3 410 user:9 # user3(250), user9(410) 한번에 추가



Clear Results				
ZADD game:ranking 250 user:3 410 user:9	Jun 23, 12:26:49	2.426 msec		
(integer) 2				
ZADD game:ranking 120 user:7	Jun 23, 12:25:13	1.946 msec		
DEL game:ranking]	Jun 23, 12:25:13	1.006 msec		

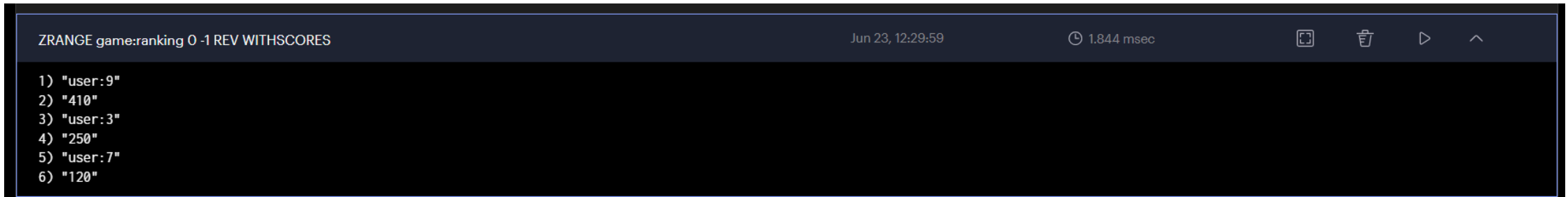
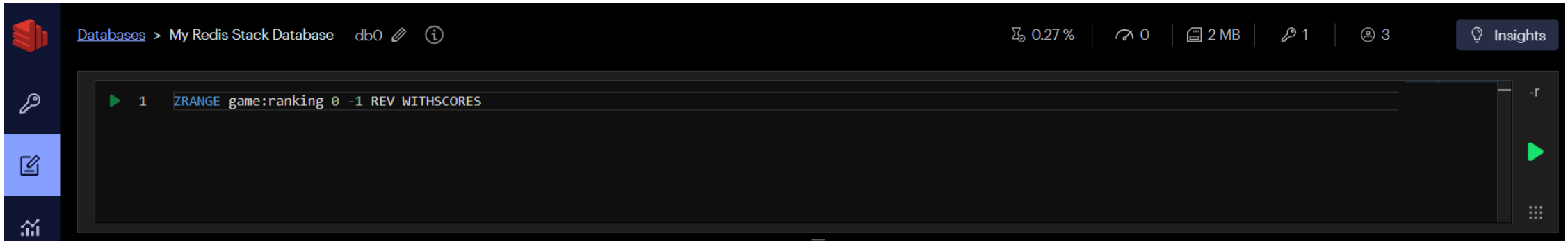
- `ZRANGE` : 낮은 점수부터 범위 조회
- `WITHSCORES` : 결과에 점수 포함

```
ZRANGE game:ranking 0 -1 WITHSCORES      # 전체 조회 (오름차순) → user7, user3, user9
```



- **REV** : 높은 점수부터 역순 조회

ZRANGE game:ranking 0 -1 REV WITHSCORES # 전체 조회 (내림차순) → user9, user3, user7



2. 점수 변경과 순위 확인

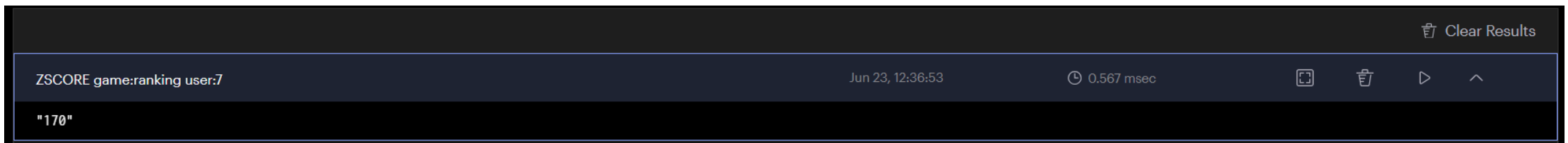
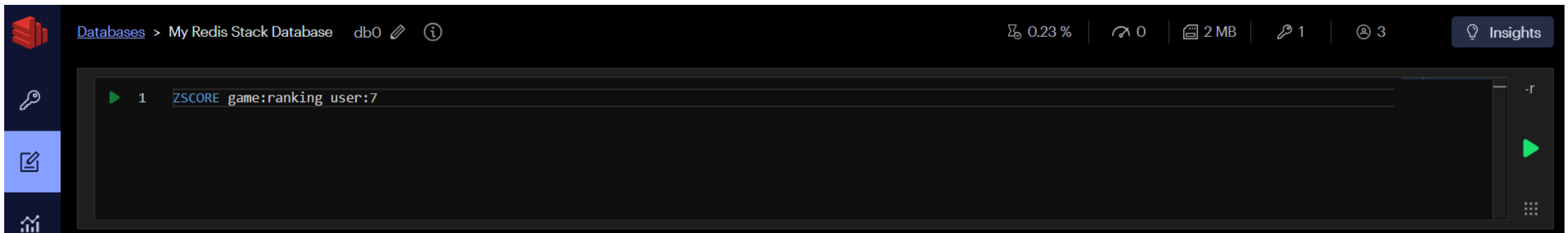
- **ZINCRBY** : Sorted Set의 점수를 지정한 값만큼 증가

ZINCRBY game:ranking 50 user:7 # user7 점수 50 증가 (120 → 170)

The screenshot shows a Redis CLI interface. The top bar displays 'Databases > My Redis Stack Database db0' and various statistics: '0.28 %', '0', '2 MB', '1', '3', and an 'Insights' button. The main command input area shows the command `ZINCRBY game:ranking 50 user:7` with a green play button on the left and a green play button on the right. Below the command input, the result of the command is displayed in a dark box: `"170"`. The result bar also shows the command `ZINCRBY game:ranking 50 user:7`, the timestamp `Jun 23, 12:36:14`, and the execution time `2.177 msec`. There are also icons for copy, paste, and other actions.

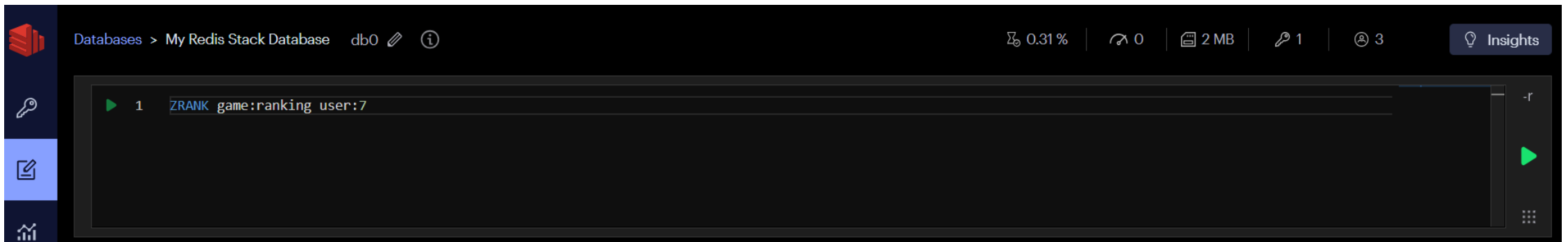
- ZSCORE : Sorted Set에서 특정 멤버의 점수를 조회

ZSCORE game:ranking user:7 # user7의 현재 점수 조회 → 170



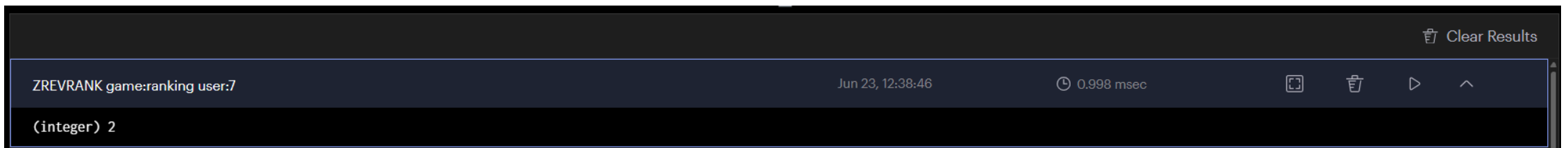
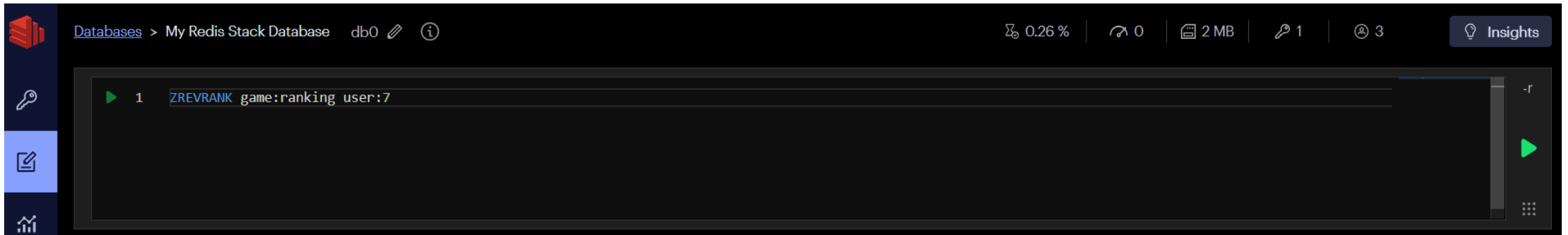
- ZRANK : 오름차순

ZRANK game:ranking user:7 # user7의 오름차순 순위 조회 → 0 (끝짜, 0부터 시작)



- ZREVRANK : 내림차순

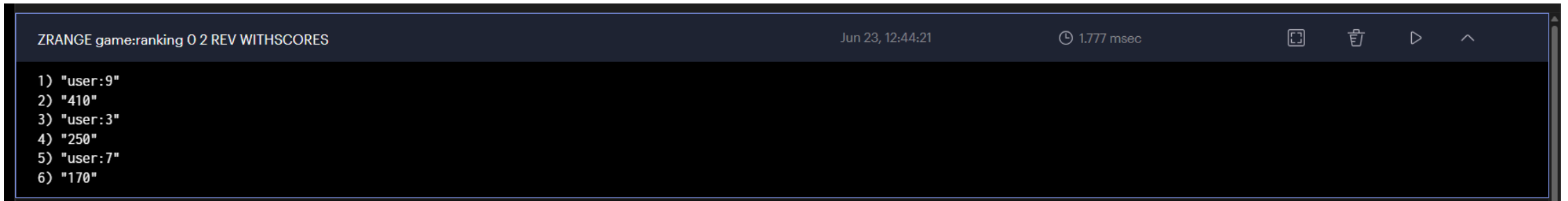
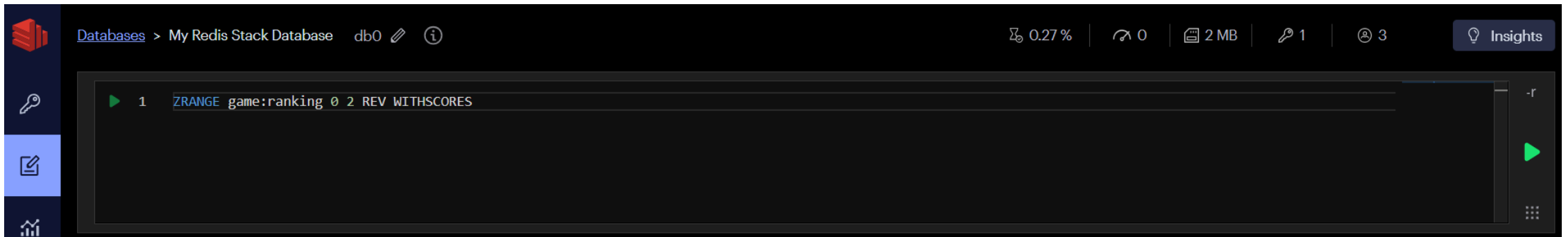
ZREVRANK game:ranking user:7 # user7의 내림차순 순위 조회 → 2 (끝찌, 0부터 시작)



3. 범위 조회

상위 세 명을 조회합니다.

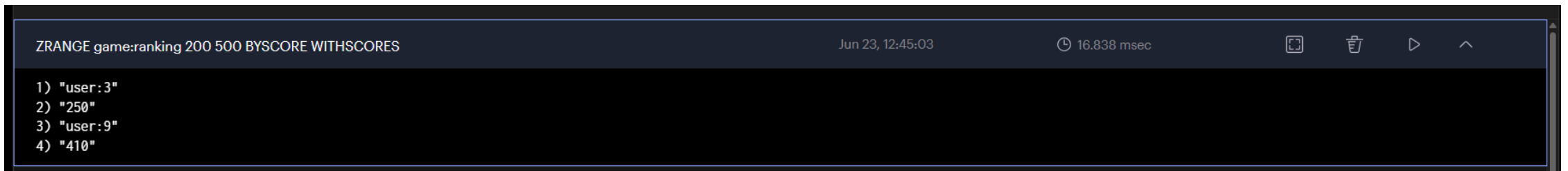
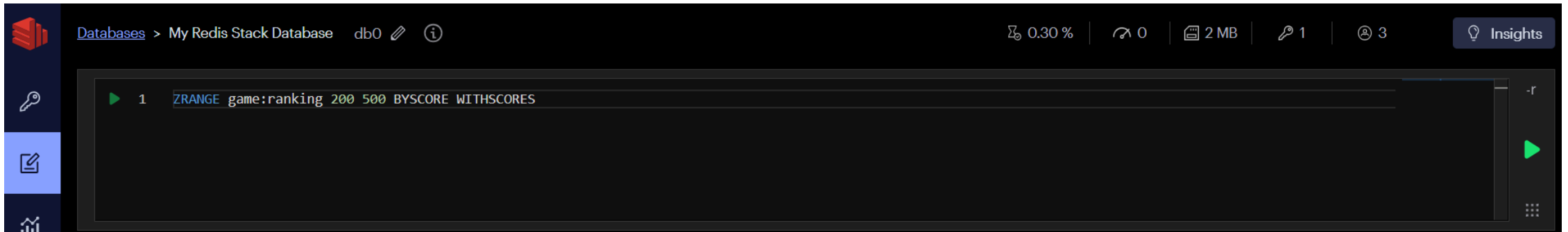
```
ZRANGE game:ranking 0 2 REV WITHSCORES
```



- BYSCORE : 점수 기준으로 범위를 지정

점수가 200 이상 500 이하인 사용자를 조회합니다.

```
ZRANGE game:ranking 200 500 BYSCORE WITHSCORES
```

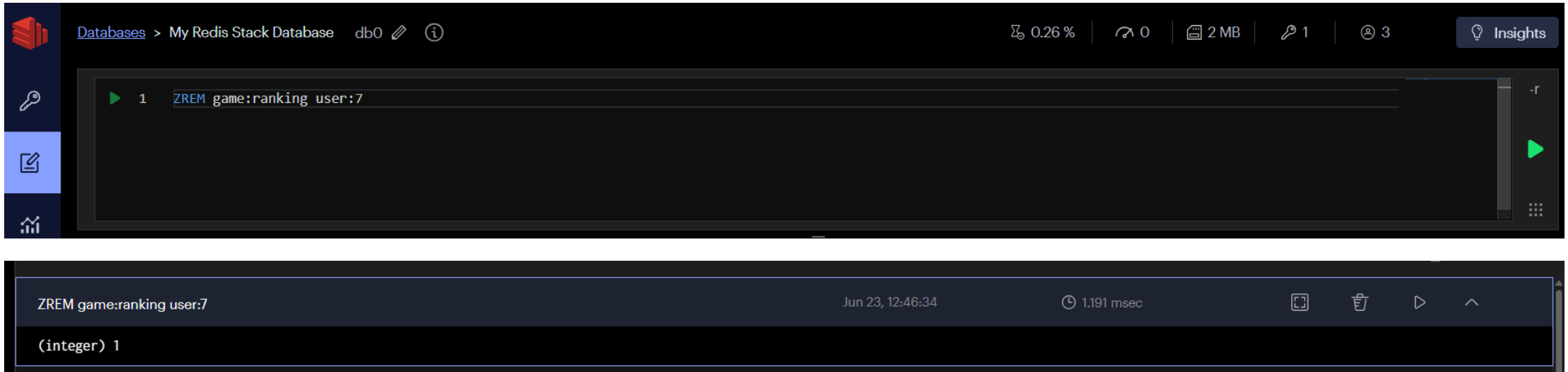


- **ZREM** : Sorted Set에서 특정 멤버를 삭제

점수나 순위 범위로 항목을 삭제할 수도 있습니다.

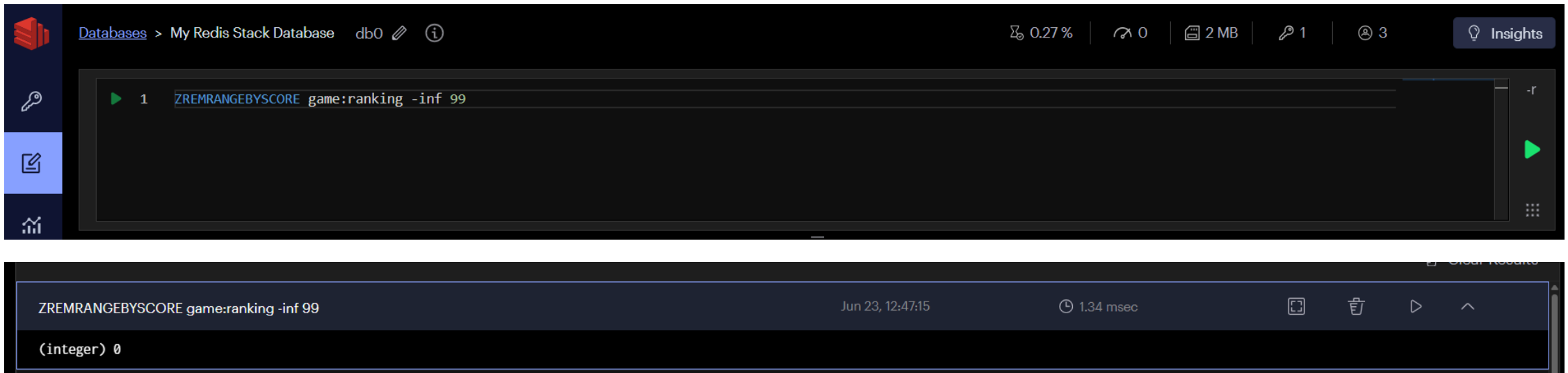
```
ZREM game:ranking user:7
```

game:ranking에서 user7 제거



- `ZREMRANGEBYSCORE` : Sorted Set에서 점수 범위에 해당하는 멤버를 일괄 삭제
점수나 순위 범위로 항목을 삭제할 수도 있습니다.

```
ZREMRANGEBYSCORE game:ranking -inf 99 # 점수 99 이하인 멤버 전체 제거
```



The screenshot shows a Redis Stack database interface. The top bar indicates the database is 'My Redis Stack Database' (db0). The command `ZREMRANGEBYSCORE game:ranking -inf 99` is entered in the command field. The interface shows the command was executed successfully, returning `(integer) 0`. The status bar at the bottom indicates the command was executed on Jun 23, 12:47:15, taking 1.34 msec.

4. 실시간 인기 게시물

- **ZINCRBY** : Sorted Set의 점수를 지정한 값만큼 증가

게시글을 조회하거나 좋아요를 누르면 점수를 높입니다.

```
ZINCRBY ranking:articles:daily 1 article:100 # article:100 조회수 1 증가 (0 → 1)
ZINCRBY ranking:articles:daily 1 article:100 # article:100 조회수 1 증가 (1 → 2)
ZINCRBY ranking:articles:daily 5 article:200 # article:200 조회수 5 증가 (0 → 5)
ZRANGE ranking:articles:daily 0 9 REV WITHSCORES # 상위 10개 조회 (내림차순) → article:200(5), article:100(2)
```



[Databases](#) > My Redis Stack Database

db0

0.30 % | 0 | 2 MB | 1 | 3

Insights

1

ZINCRBY ranking:articles:daily 1 article:100

2

ZINCRBY ranking:articles:daily 1 article:100

3

ZINCRBY ranking:articles:daily 5 article:200

4

ZRANGE ranking:articles:daily 0 9 REV WITHSCORES

5

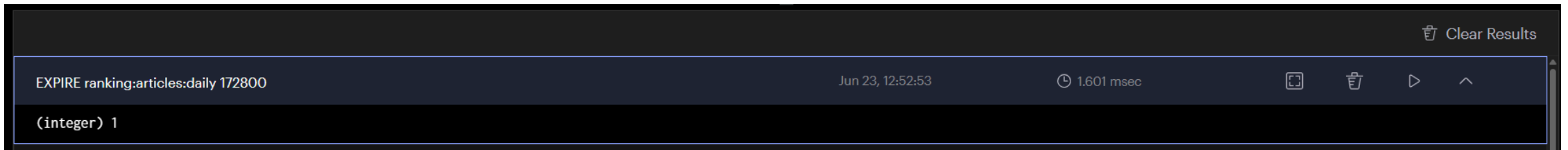
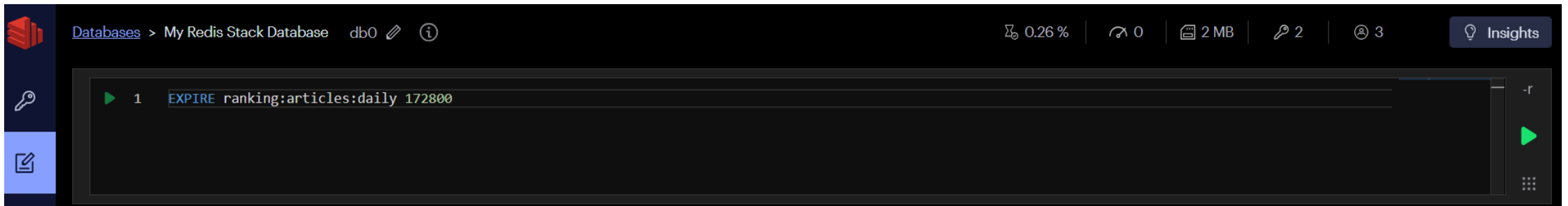
-r

ZRANGE ranking:articles:daily 0 9 REV WITHSCORES	Jun 23, 12:51:11	3.489 msec				
1) "article:200"						
2) "5"						
3) "article:100"						
4) "2"						
ZINCRBY ranking:articles:daily 5 article:200	Jun 23, 12:51:11	3.547 msec				
"5"						
ZINCRBY ranking:articles:daily 1 article:100	Jun 23, 12:51:11	3.66 msec				
"2"						
ZINCRBY ranking:articles:daily 1 article:100	Jun 23, 12:51:11	1.352 msec				
"1"						

- **EXPIRE** : 키에 만료 시간(초) 을 설정

일별 랭킹 키에 TTL을 설정하면 오래된 랭킹을 자동으로 정리할 수 있습니다.

```
EXPIRE ranking:articles:daily 172800 # ranking:articles:daily를 172800초(48시간=2일) 후 자동 삭제
```



5. List, Set, Sorted Set 비교

질문	적합한 자료구조
입력된 순서가 중요한가?	List
중복 없이 포함 여부만 필요한가?	Set
중복 없이 점수와 순위가 필요한가?	Sorted Set